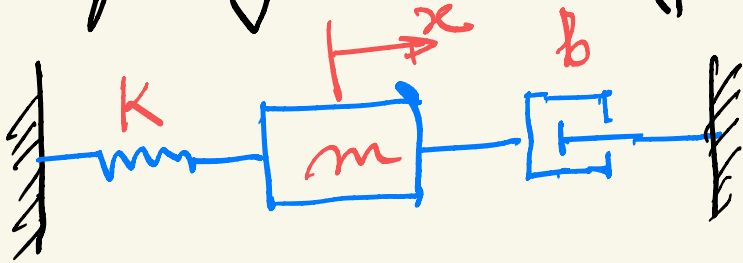




Lecture 3

In the last class we have derived the differential equation of a spring-mass system without damping. In this class we will add an extra term that accounts for friction and damping. Note that many different physical systems can be realized or modeled with this spring mass system. As for example, you can represent a reciprocating or rotary

motion of an engine with
a spring mass system.



$$\text{spring force} = -Kx$$

$$\begin{aligned}\text{damping force} &= -b\dot{x} \\ &= -b \frac{dx}{dt}\end{aligned}$$

$$\vec{+} \sum F_x = ma$$

$$\Rightarrow -Kx - b \frac{dx}{dt} = m \frac{d^2x}{dt^2} \quad \text{--- (1)}$$

Note that x is a function of
time, that is $x(t)$ changes with
time.

By rearranging equation ① we get

$$m \frac{d^2 x(t)}{dt^2} + b \frac{dx(t)}{dt} + k x(t) = 0 \quad \text{--- ②}$$

Note that the right hand side of equation ② is zero. This means that no external force has been applied on the system.

Equation ② has a specific name. It is referred to as

2nd order constant coefficient
homogeneous differential equation.

m, b, k are constant coefficients or parameters

$\frac{d^2 x(t)}{dt^2}$ has 2nd order derivative.

This is also a linear differential equation.

* There is no non-linear term in this equation

let's assume that the solution is in the form $x(t) = e^{at}$

$$\frac{dx(t)}{dt} = a e^{at}$$

$$\frac{d^2 x(t)}{dt^2} = a^2 e^{at}$$

plugging this in equation ② we obtain the following equation

$$m \cdot a^2 e^{at} + b \cdot a e^{at} + k \cdot e^{at} = 0$$

$$\Rightarrow e^{at} (ma^2 + ba + k) = 0$$

$e^{at} \neq 0$, as a result

$$ma^2 + ba + k = 0$$

This is a quadratic equation in a . The two roots are as follows

$$ma^2 + ba + k = 0$$

$$\Rightarrow a_{1,2} = \frac{-b \pm \sqrt{b^2 - 4mk}}{2m}$$

The two solution is

$$x_1(t) = C_1 e^{a_1 t}$$

$$x_2(t) = C_2 e^{a_2 t}$$

where C_1 and C_2 is arbitrary constants.

The general solution of equation (2) is

$$x(t) = C_1 e^{a_1 t} + C_2 e^{a_2 t}$$

In order to find the value of C_1 and C_2 , two initial condition is needed which is problem specific.

Let's say, initial displacement, that is at $t=0$, $x(0) = 20$ cm and initial velocity $\frac{dx(0)}{dt} = 0$.

We will use this two initial

conditions to find the value of c_1 and c_2 .

$$x(0) = c_1 e^{a_1 \cdot 0} + c_2 e^{a_2 \cdot 0} = 20$$

$$\Rightarrow c_1 \cdot 1 + c_2 \cdot 1 = 20$$

$$\Rightarrow \boxed{c_1 + c_2 = 20} \Rightarrow c_1 = 20 - c_2$$

$$\frac{dx(0)}{dt} = c_1 a_1 e^{a_1 \cdot 0} + c_2 a_2 e^{a_2 \cdot 0} = 0$$

$$\Rightarrow \boxed{c_1 a_1 + c_2 a_2 = 0}$$

$$\Rightarrow (20 - c_2) a_1 + c_2 a_2 = 0$$

$$\Rightarrow 20 a_1 - c_2 a_1 + c_2 a_2 = 0$$

$$\Rightarrow 20 a_1 + c_2 (a_2 - a_1) = 0$$

$$\Rightarrow c_2 (a_2 - a_1) = -20 a_1$$

$$\Rightarrow C_2 = \frac{-20a_1}{a_2 - a_1}$$

$$C_1 = 20 - C_2$$

$$= 20 + \frac{20a_1}{a_2 - a_1}$$

$$= \frac{20a_2 - \cancel{20a_1} + \cancel{20a_1}}{a_2 - a_1}$$

$$C_1 = \frac{20a_2}{a_2 - a_1}$$

As a result, we have found the value of two constants.

$$C_1 = \frac{20a_2}{a_2 - a_1}$$

$$C_2 = -\frac{20a_1}{a_2 - a_1}$$

Now putting this value into the solution

$$x(t) = C_1 e^{a_1 t} + C_2 e^{a_2 t}$$

$$x(t) = \frac{20a_2}{a_2 - a_1} \cdot e^{a_1 t} - \frac{20a_1}{a_2 - a_1} \cdot e^{a_2 t}$$

This is how we found the complete solution. This type of diffe-

differential equations are also called initial value problems. we have found an analytical solution of the differential equation that we formulated.